

AUTUMN (GAEUL) KWON

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Summary

Research-focused **Physical AI engineer** with experiences in **reasoning VLA** models and publications in **in-context learning**. Ongoing work on **continual learning** with VLA models for **robot manipulation** and **sim-to-real transfer**.

Technical Skills

Programming & Frameworks: Python, C, PyTorch, JAX, Hugging Face Transformers

Machine Learning: LLMs, VLMs, VLA Models, Reinforcement Learning (RL), Imitation Learning (IL)

Robotics & Simulation: Robot Manipulation, Sim-to-Real Transfer, Domain Randomization (DR), Isaac Gym

Tools & Infrastructure: Git, Linux, SLURM, GPU Computing

Education

University of Southern California

Master of Science in Computer Science

Los Angeles, California

Aug 2025 - May 2027

Grinnell College

Bachelor of Arts in Computer Science (Minor in Linguistics)

Grinnell, Iowa

Aug 2019 - May 2025

Research Experience

Sensing, Learning, and Understanding for Robotic Manipulation Lab, USC

Graduate Research Assistant

Los Angeles, California

Oct 2025 - Present

- Investigate **world model (WM)**-based verification for reasoning vision-language-action (VLA) policies, evaluating action-selection reliability and generalization on out-of-distribution (OOD) robotic manipulation tasks.
- Develop **PyTorch**-based training and evaluation pipelines for reasoning VLA policies in simulation, supporting systematic benchmarking across models and task conditions.

University of Illinois Urbana-Champaign

Undergraduate Research Assistant

Remote

Mar 2024 - Apr 2025

- Designed a comparative prompting framework for **in-context learning** using **PyTorch** and **Hugging Face Transformers**, systematically analyzing prompt bias and generalization failures in large language models.
- Achieved up to **13.8%** accuracy improvement across **LLaMA 2/3** and **Flan-T5 XL/XXL**, enabling smaller models to match or exceed larger-model baselines.
- Identified **sparse autoencoder (SAE) reconstruction loss** as a strong unsupervised signal for prompt selection, extending SAE usage from interpretability to controllable model steering.

AI and Optimization Lab (PiStar), Texas A&M University

Undergraduate Researcher

College Station, Texas

May 2024 - Aug 2024

- Leveraged **curriculum learning to reinforcement learning (RL)** policies for autonomous racing using vision-based observations, trained entirely in simulation.
- Implemented visual **domain randomization** (lighting direction, intensity, shadows) in **PyTorch** to improve **sim-to-real** robustness.
- Trained **autoencoder**-based visual representations to enhance feature transferability between simulated and real-world environments.

Industry Experience

StudyFetch, LLC.

AI Research Assistant – Machine Learning & AI

Beverly Hills, California

Oct 2026 - Present

- Develop evaluation and benchmarking pipelines for AI-powered educational applications, assessing model performance across educational datasets and K-12 learning interactions.
- Design and analyze model evaluation methodologies to identify performance limitations and improve the reliability of AI-generated educational responses.

PiQuant Co., Ltd.

Firmware Developer

Seoul, South Korea

Jun 2022 - Mar 2023

- Developed and deployed firmware for a portable spectroscopic analyzer, improving device reliability and measurement stability.
- Implemented low-level control logic and hardware communication in **C** on embedded systems.

Projects

Safe Dexterous Manipulation with ORCA Hand

Fall 2026

Imitation and Reinforcement Learning, Multi-Step Robot Manipulation

- Developing a multi-step robotic manipulation framework using an ORCA dexterous hand, combining **imitation learning (IL)** for pick-and-place with **reinforcement learning (RL)** for contact-rich tool manipulation.
- Designing a safety-aware control pipeline with motion constraints, collision checks, and RGB-based progress verification to support reliable sim-to-real transfer.

Reinforcement Learning (RL) with Symbolic Feedback for Physics-Based Motion Imitation

Spring 2026

Reinforcement Learning, Neurosymbolic Feedback

- Augmented the vanilla **reinforcement learning** algorithms (PPO and AWR) with interpretable symbolic signals to improve training stability, sample efficiency, and motion quality through structured feedback.
- Constructed an end-to-end **symbolic feedback** pipeline with **JAX** in **Issac Gym** environment.

Large Language Model (LLM) Sycophancy in Multi-Turn Reasoning Dialogues

Spring 2026

LLM Sycophancy, LLM Reasoning

- Investigated robustness of LLMs to **sycophancy** in multi-turn dialogues, focusing on the effects of reasoning and user authority.
- Constructed trajectory-based metrics to compare response dynamics of instruction-tuned and reasoning-oriented models under controlled dialogue settings.

Publications

- [1] Ikhyun Cho, **Gaeul Kwon**, and Julia Hockenmaier. “On the Versatility of Sparse Autoencoders for In-Context Learning”. *Findings of the Association for Computational Linguistics: EMNLP 2025*. URL: <https://aclanthology.org/2025.findings-emnlp.1063/>.
- [2] Ikhyun Cho, **Gaeul Kwon**, and Julia Hockenmaier. “Tutor-ICL: Guiding Large Language Models for Improved In-Context Learning Performance”. *Findings of the Association for Computational Linguistics: EMNLP 2024*. URL: <https://aclanthology.org/2024.findings-emnlp.554>.